

Customer Shopping Behavior Analysis

Uncovering actionable insights from 3,900 customer purchases to drive strategic business growth and optimize revenue streams



Dataset Overview

3,900

Total Purchases

Comprehensive
transaction dataset
analyzed

18

Data Columns

Rich customer
demographics and
behavioral attributes

37

Missing Values

Minimal gaps—only
in Review Rating
column

Data Preparation Process

01

Data Loading

Imported complete dataset using pandas framework

02

Initial Exploration

Structure validation and summary statistics review

03

Missing Data Handling

Review Rating gaps filled with median imputation

04

Feature Engineering

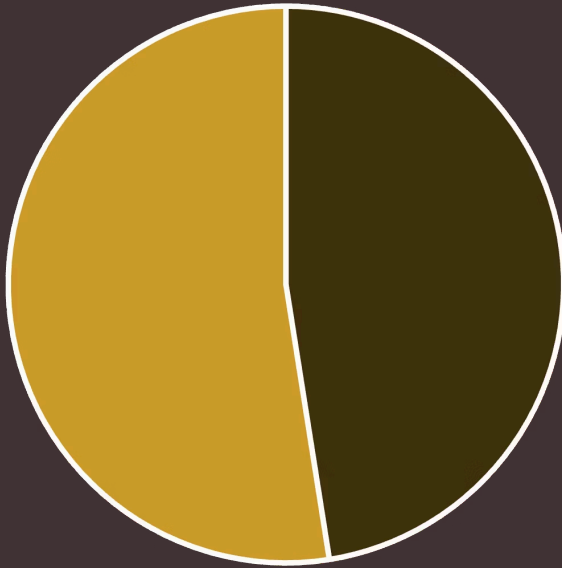
Created age groups and purchase frequency segments

05

Database Integration

Connected to PostgreSQL for advanced analysis

Revenue by Gender Analysis



Male vs Female customer revenue distribution

Key Insight

Female customers generate slightly higher total revenue than male customers—opening opportunities for gender-based marketing strategies to optimize conversion and lifetime value.

- 📌 **Action:** Tailor campaigns by gender preferences to maximize revenue potential across segments

High-Value Discount Users

Identifying premium customers who spend above average while actively utilizing discount offers

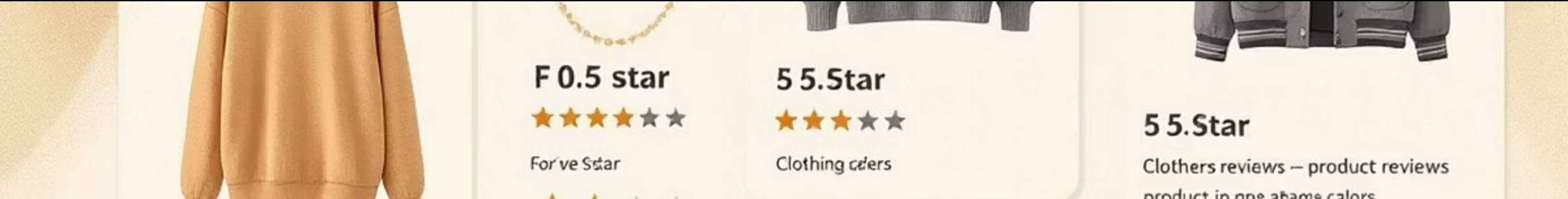
Smart Shoppers

High-spending customers who maximize value through strategic discount usage—representing our most price-conscious premium segment

Strategic Opportunity

Target these savvy buyers with exclusive VIP offers, early access promotions, and tiered loyalty rewards to drive retention and increase purchase frequency





Top-Rated Products

Blouse ★ ★ ★ ★ ★ 5 Highest customer satisfaction rating	Dress ★ ★ ★ ★ ★ 5 Consistently excellent customer reviews	Shirt ★ ★ ★ ★ ★ 4 Strong customer approval rating
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These top performers represent the foundation of our product strategy—prioritize inventory, feature prominently in campaigns, and use as benchmarks for new product development

Shipping Preferences Impact



Express Shipping

\$65

Average purchase amount per transaction



Standard Shipping

\$58

Average purchase amount per transaction

Express shipping customers spend 12% more per transaction—indicating higher urgency and willingness to pay premium prices



Subscription Impact



Higher Spend

Subscribers consistently outspend non-subscribers across all categories



Revenue Share

Disproportionate revenue contribution from subscriber base

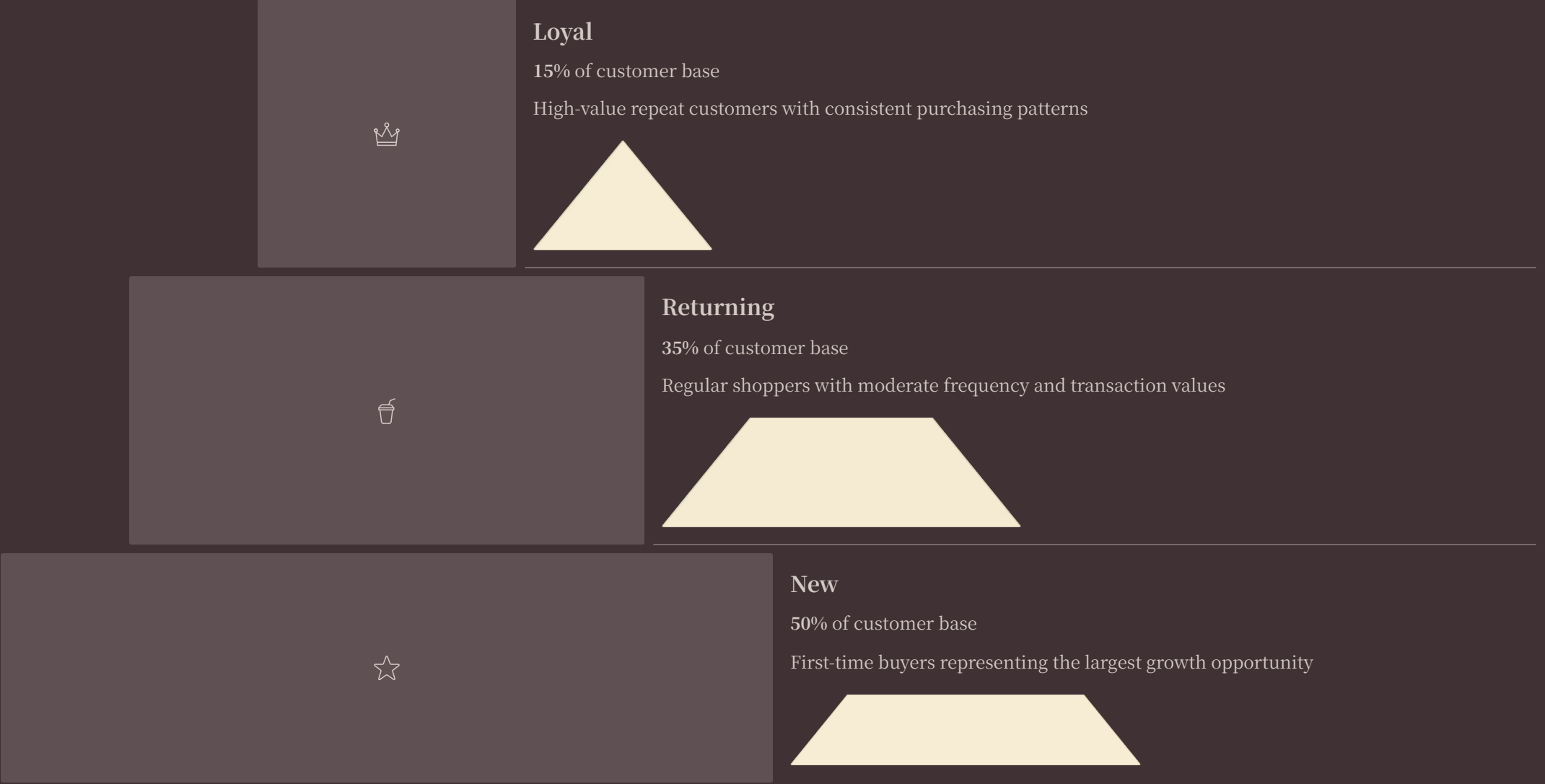


Loyalty Rate

Significantly higher repeat purchase frequency among subscribers

Bottom Line: Subscription customers are the most valuable segment—driving recurring revenue, higher transaction values, and exceptional lifetime value

Customer Segmentation



📌 **Strategic Focus:** Convert New customers to Returning status, then nurture Returning customers into Loyal advocates through targeted engagement and personalized experiences



Strategic Recommendations

→ Boost Subscriptions

Promote exclusive benefits, early access, and VIP perks to drive subscription adoption and increase customer lifetime value

→ Loyalty Programs

Implement tiered reward systems that incentivize repeat purchases and move customers up the loyalty ladder from New to Returning to Loyal

→ Targeted Marketing

Focus campaigns on high-revenue segments including express shipping users, female customers, and discount-savvy premium buyers

→ Product Positioning

Feature top-rated products (Blouse, Dress, Shirt) prominently in campaigns, inventory planning, and cross-sell recommendations